THE NATIONAL INSTITUTE OF ENGINEERING

Mysuru – 570 008

VIRTUAL LAB REPORT

COURSE: Linear Integrated Circuits

COURSE CODE : BEC405D

Submitted By:

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Experiment:

Measurement of Input and Output Offset Voltage and Slew Rate of IC 741

Aim:

To measure the Input Bias Current, Input Offset Current, Input Offset Voltage, Output Offset Voltage and Slew Rate of IC 741 using the virtual lab.

Apparatus Required:

• IC 741
• DC Power Supply (+Vcc and -Vcc)
• Multimeter
• Function Generator
• Connecting Wires

Theory:

An ideal op-amp draws no current at its input terminals and has zero offset voltage. However, practical op-amps, such as the IC 741, have non-ideal characteristics, including Input Bias Current, Input Offset Current, Input Offset Voltage, Output Offset Voltage, and Slew Rate.

- Input Bias Current (I_B): Average of currents entering inverting and non-inverting terminals.

$$I_B = \frac{(IB1 + IB2)}{2}$$

- Input Offset Current (I_{OS}): Difference between IB1 and IB2.

$$I_{OS} = |I_{B1} - I_{B2}|$$

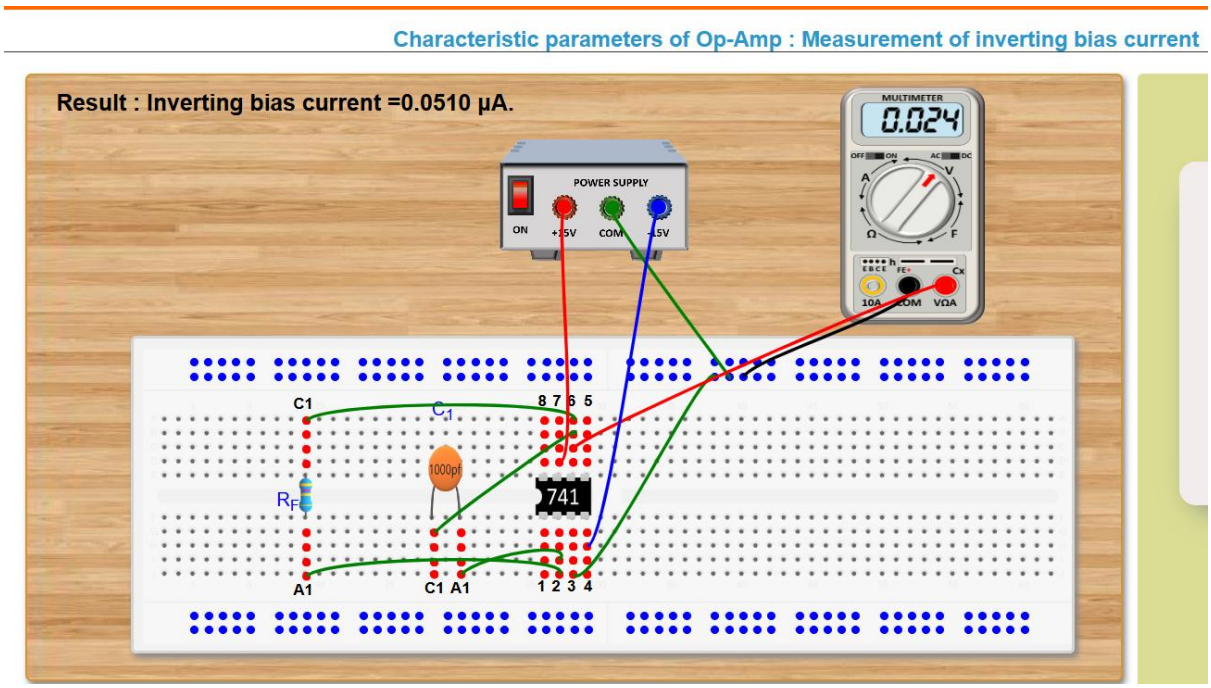
- Input Offset Voltage (V_{IO}): Small differential voltage required to make the output zero.
- Output Offset Voltage: Output voltage present when the input is zero.
- Slew Rate (SR): Maximum rate of change of output voltage.

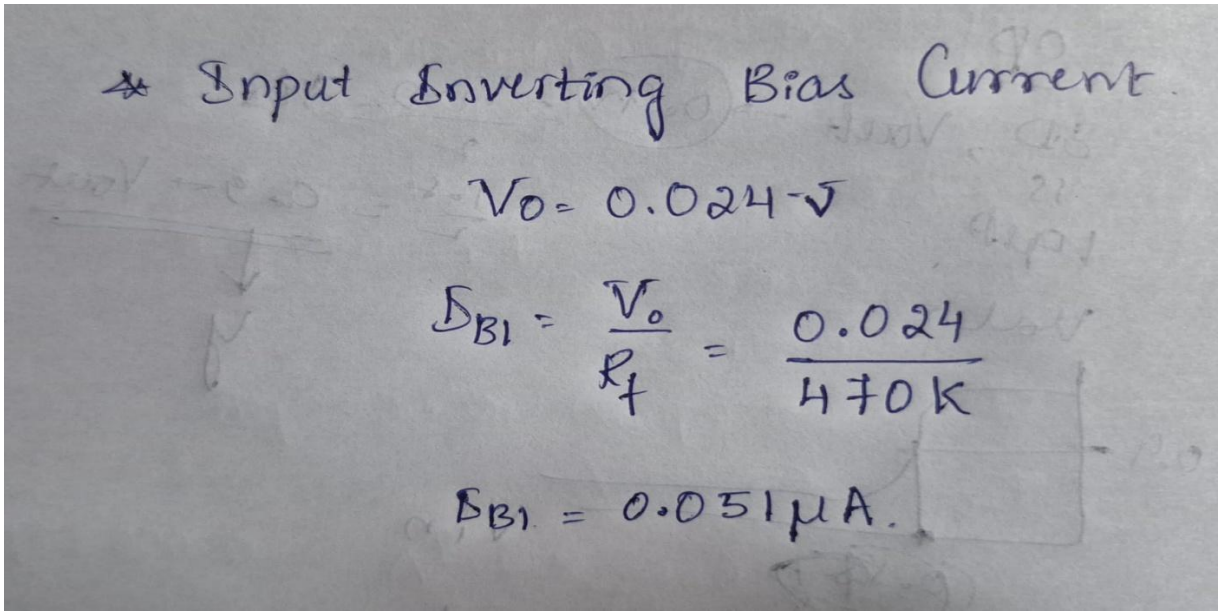
$$SR = \left(\frac{dVo}{dt}\right)_{max} \quad (V/\mu s)$$

Circuit Diagrams and Calculations:

1) Input Bias Current

a) Input Inverting Bias Current





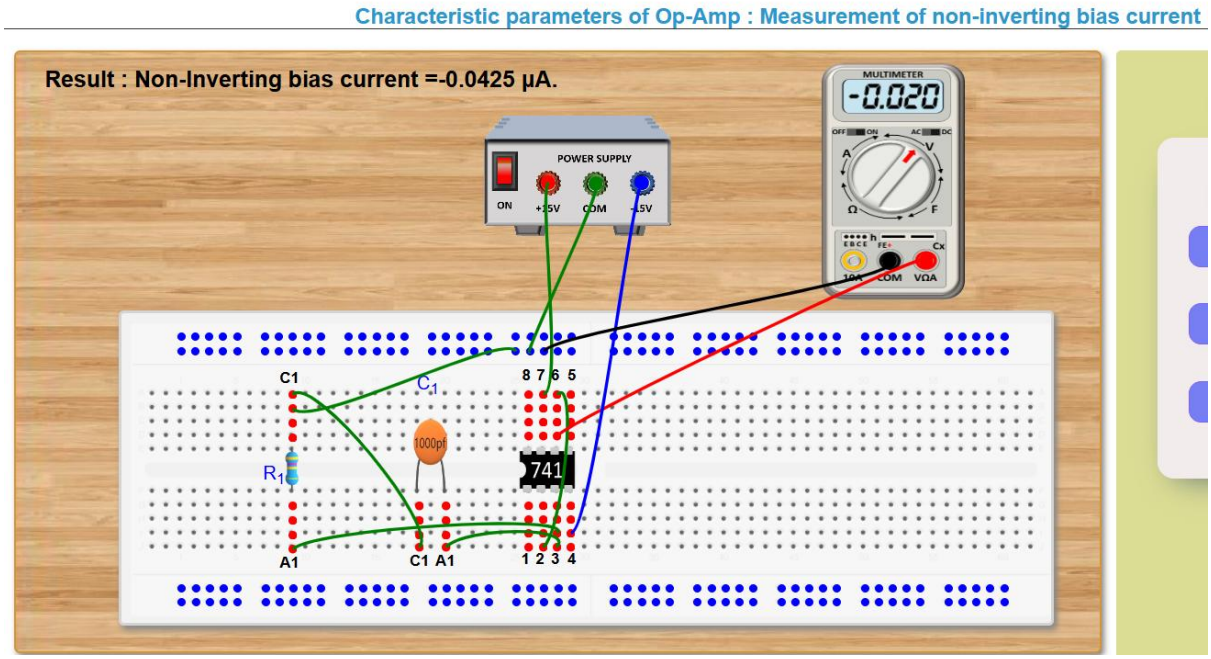
* Input Inverting Bias Current

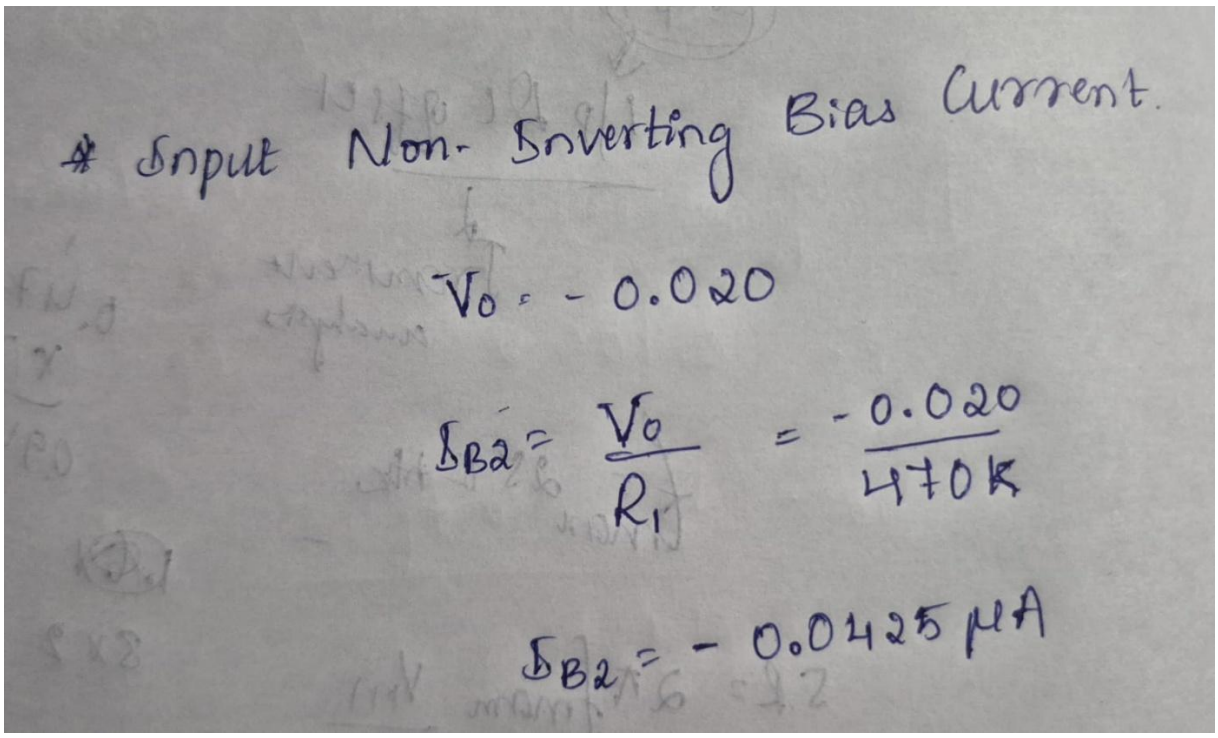
$V_0 = 0.024\,V$

$I_{B1} = \frac{V_0}{R_f} = \frac{0.024}{470K}$

$I_{B1} = 0.051\,\mu A.$

b) Input Non - Inverting Bias Current





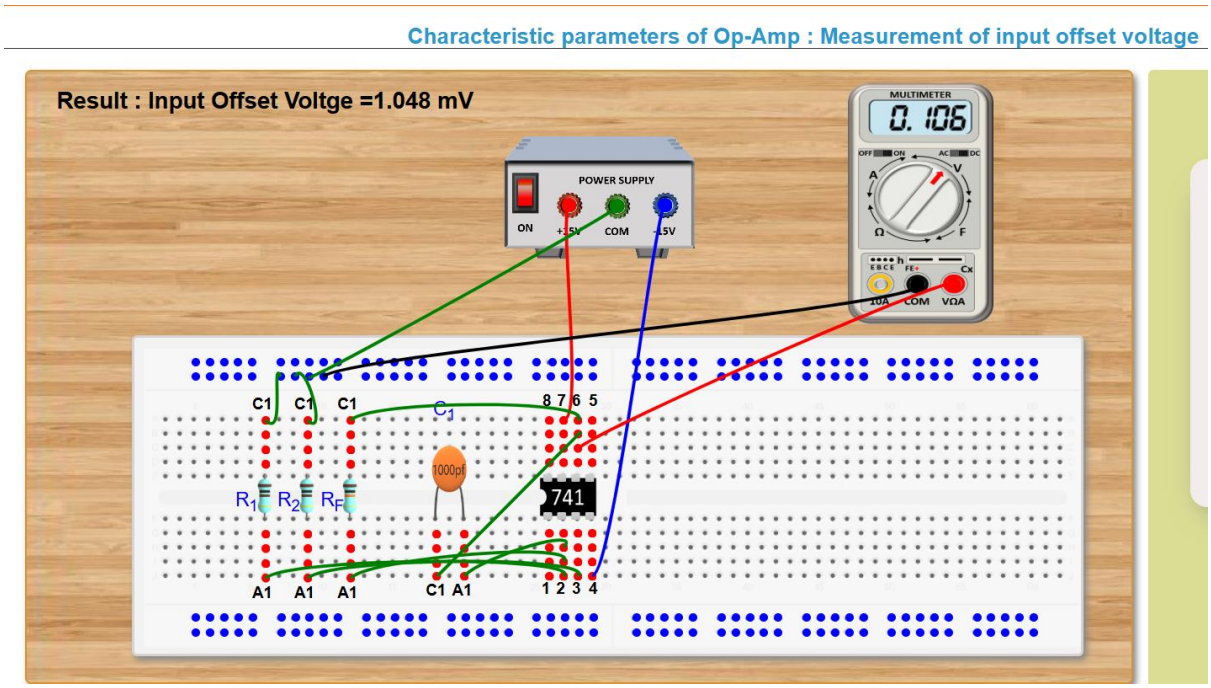
* Input Non- Inverting Bias Current.

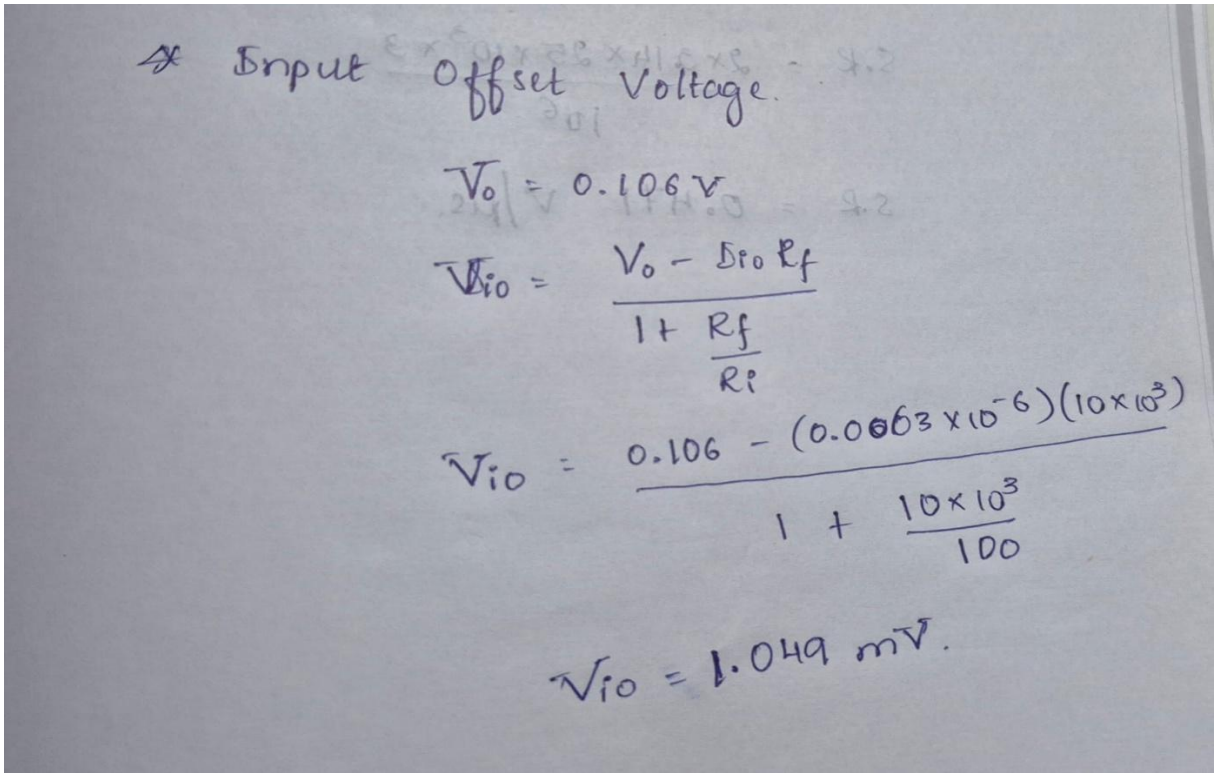
$V_o = -0.020$

$I_{B2} = \frac{V_o}{R_1} = \frac{-0.020}{470K}$

$I_{B2} = -0.0425\ \mu A$

2) Input Offset Voltage





* Input Offset Voltage.

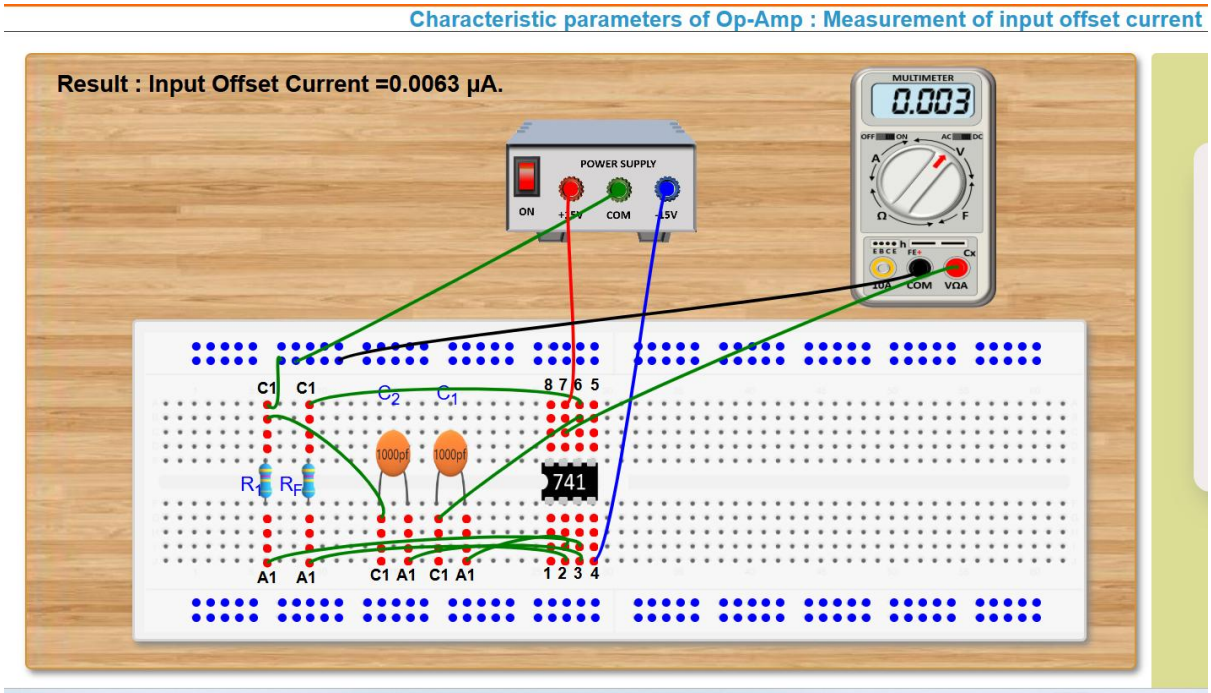
$V_o = 0.106\,V$

$$V_{io} = \frac{V_o - I_{io} R_f}{1 + \frac{R_f}{R_i}}$$

$$V_{io} = \frac{0.106 - (0.0063 \times 10^{-6})(10 \times 10^{3})}{1 + \frac{10 \times 10^{3}}{100}}$$

$V_{io} = 1.049\ mV.$

3) Input Offset Current :



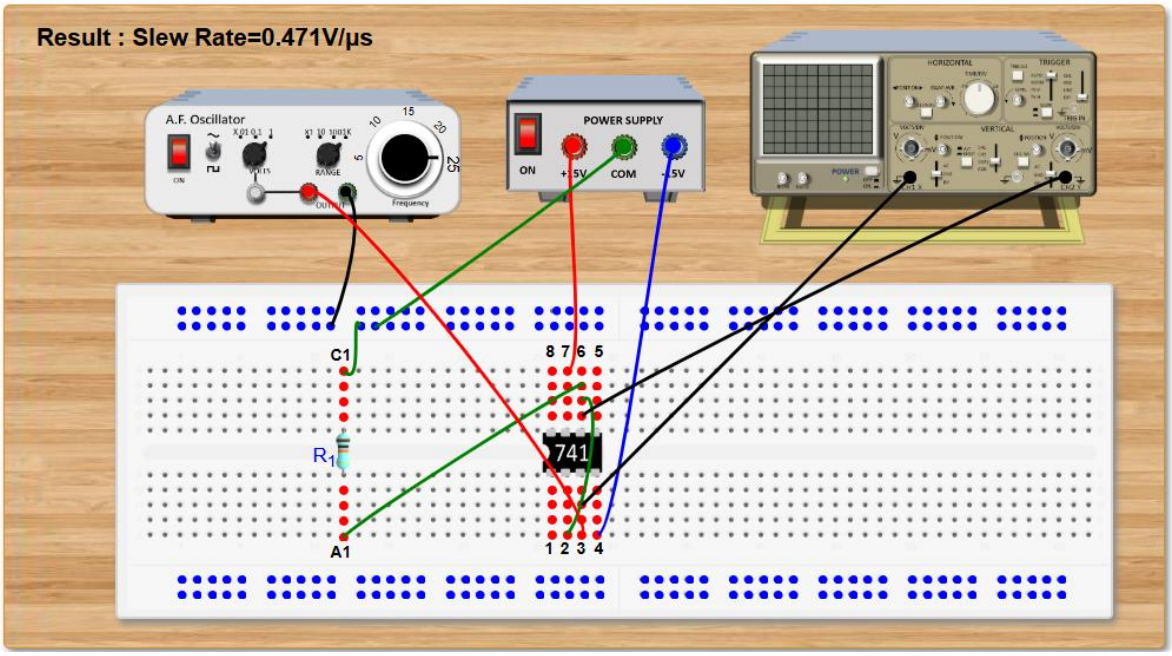
* Input Offset Current

$V_o = 0.003V$

$I_{io} = \frac{V_o}{R_f} = \frac{0.003}{470k}$

$I_{io} = 0.0063\mu A$

4) Slew Rate



* Slew Rate.

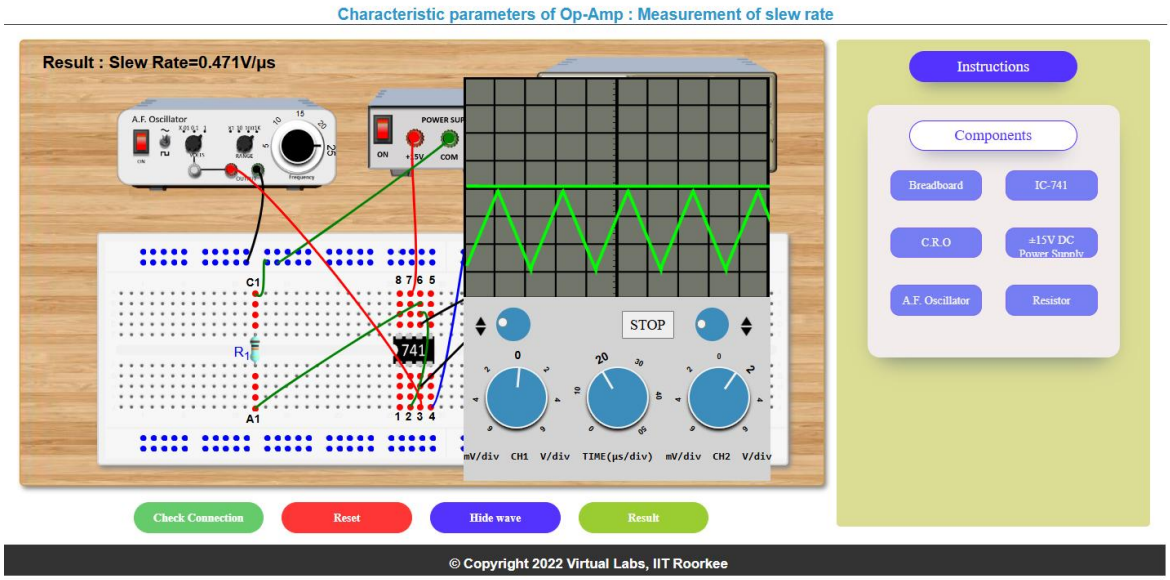
$f_{max} = 25\,KHz.$

$V_m = 3$

$$S.R = \frac{2\pi f_{max} \cdot V_m}{10^6} \quad V/\mu s.$$

$$S.R = \frac{2 \times 3.14 \times 25 \times 10^3 \times 3}{10^6}$$

$$S.R = 0.471 \quad V/\mu s.$$



Observation Table :

Sl. No	Parameter	Measured Value
1	Input Bias Current	Inverting - 0.051 uA
		Non-inverting - -0.0425 uA
2	Input Offset Current	0.0063 uA
3	Input Offset Voltage	1.049 mV
4	Slew Rate	0.471 V/us

Result:

The parameters of the IC 741 were successfully measured.

Conclusion:

Thus, the non-ideal characteristics of IC 741 such as input bias current, offset current, offset voltage and slew rate were studied and verified.